IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- Summary of methodologies
  - The following methodologies are applied
    - Data wrangling on public data
    - Exploratory data anslysis using both visualization and SQL
    - Interactive visual analytics and dashboards
    - Predative analysis on classification
- Summary of all results
  - The following results are included in this pack
    - The data frame after data wrangling techniques are applied
    - The different plot and query results
    - The interactive map and dashboard
    - The results of different predictive method and the evaluation of the accuracy of respective method

Introduction

- Project background and context
  - To achieve cost saving if first stage can be reused
  - Train machine learning model and use public information to predict if SpaceX will reuse the first stage.
- Problems you want to find answers
  - To determine the price of each launch.

Section 1

Methodology

Methodology

Executive Summary

- Data collection methodology:
  - Describe how data was collected
- Perform data wrangling
  - Describe how data was processed
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
  - How to build, tune, evaluate classification models

Data Collection

- Describe how data sets were collected.
- You need to present your data collection process use key phrases and flowcharts

Data Collection – SpaceX API

- Present your data collection with SpaceX REST calls using key phrases and flowcharts

- Add the GitHub URL of the completed SpaceX API calls notebook (must include completed code cell and outcome cell), as an external reference and peer-review purpose

1. Use GET request to request the data from URL: requests.get(static_json_url)

-> 2. Decode the response content and convert to Pandas dataframes: pd.json_normalize(response.json())

-> 3. Extract key attributes and combine to a dictionary

GitHub URL: https://github.com/lftsang25/Final_Project/blob/main/jupyter-labs-spacex-data-collection-api_completed.ipynb

Data Collection - Scraping

- Present your web scraping process using key phrases and flowcharts

- Add the GitHub URL of the completed web scraping notebook, as an external reference and peer-review purpose

1. Request launch data from Wiki page via URL: requests.get(static_url)

-> 2. Extract and parse the data and store them in Data Frame

GitHub URL:

https://github.com/lftsang25/Final_Project/blob/main/jupyter-labs-webscraping_completed.ipynb

Data Wrangling

- Describe how data were processed
  - Apply exploratory data analysis and determine training labels
- You need to present your data wrangling process using key phrases and flowcharts
- Add the GitHub URL of your completed data wrangling related notebooks, as an external reference and peer-review purpose
  - GitHub URL: https://github.com/lftsang25/Final_Project/blob/main/labs-jupyter-spacex-Data%20wrangling_completed.ipynb

EDA with Data Visualization

- Summarize what charts were plotted and why you used those charts
  - Scatter plot is used to visualize the relationship between Flight Number and Launch Site
  - Scatter plot is used to visualize the relationship between Payload Mass and Launch Site
  - Bar chart is used to visualize the relationship between Success Rate of each Orbit type
  - Scatter plot is used to visualize the relationship between Flight Number and Orbit type
  - Scatter plot is used to visualize the relationship between Payload Mass and Orbit Type
  - Line plot is used to visualize the yearly trend of Success Rate
- Add the GitHub URL of your completed EDA with data visualization notebook, as an external reference and peer-review purpose
  - GitHub URL: https://github.com/lftsang25/Final_Project/blob/main/edadataviz_Completed.ipynb

EDA with SQL

- Using bullet point format, summarize the SQL queries you performed
  - Select and display the names of the unique launch sites
  - Select and display the 5 records with launch site starting with CCA
  - Select and display the total payload mass carried by boosters launched by NASA (CRS)
  - Select and display the average payload mass carried by booster version F9 v1.1
  - Select and display the date when the first successful landing outcome in ground pad was achieved
  - Select and display the names of the boosters which have success in drone ship with payload mass between 4000 and 6000
  - Select and display the total number of success and failure outcomes
  - Select and display the booster version that have carried the maximum payload mass
  - Select and display the month, failure landing outcome, booster version and launch site for year 2015
  - Select and display the count of landing outcome between 2016-06-04 and 2017-03-20
- Add the GitHub URL of your completed EDA with SQL notebook, as an external reference and peer-review purpose
  - GitHub URL: https://github.com/lftsang25/Final_Project/blob/main/jupyter-labs-eda-sql-coursera_sqllite_Completed.ipynb

Build an Interactive Map with Folium

- Summarize what map objects such as markers, circles, lines, etc. you created and added to a folium map
  - folium.map.marker, folium.Circle, MarkerCluster, folium.PolyLine objects are used
- Explain why you added those objects
  - folium.map.marker is used to create an icon as text label on the map
  - folium.Circle is used to add highlighted circle area with a text label on a specific coordinate
  - MarkerCluster is used to highlight different area with success or failure with different colored marker
  - Folium.PolyLine is used to show the distance between the different coordinates on the map
- Add the GitHub URL of your completed interactive map with Folium map, as an external reference and peer-review purpose
  - GitHub URL: https://github.com/lftsang25/Final_Project/blob/main/lab_jupyter_launch_site_location_Completed.ipynb

Build a Dashboard with Plotly Dash

- Summarize what plots/graphs and interactions you have added to a dashboard
- Explain why you added those plots and interactions
- Add the GitHub URL of your completed Plotly Dash lab, as an external reference and peer-review purpose

Predictive Analysis (Classification)

- Summarize how you built, evaluated, improved, and found the best performing classification model
  1. Standardize the data, split the data into training and testing set
  2. Based on the training set to build different model and use the testing set to evaluate the score of the respective model
  3. Examine the score to identify the best performing classification model
- You need present your model development process using key phrases and flowchart
  1. Load and standardized the data frame, then split the data into training and testing data set use train_test_split function
  2. Create regression and GridSearchCV object, then fit the object to find the best parameters from configured dictionary
  3. Calculate the accuracy on the test data using the score method and examine the confusion matrix
  4. Create a support vector machine and GridSearchCV object, then fit the object to find the best parameter from configured dictionary
  5. Calculate the accuracy on the test data using the score method and examine the confusion matrix
  6. Create a decision tree classifier and GridSearchCV object, then fit the object to find the best parameter from configured dictionary
  7. Calculate the accuracy on the test data using the score method and examine the confusion matrix
  8. Create a k0nearest neighbor and GridSearchCV object, then fit the object to find the best parameter from configured dictionary
  9. Calculate the accuracy on the test data using the score method and examine the confusion matrix
- Add the GitHub URL of your completed predictive analysis lab, as an external reference and peer-review purpose
  - GitHub URL: https://github.com/lftsang25/Final_Project/blob/main/SpaceX_Machine%20Learning%20Prediction_Part_5_Completed.ipynb

Results

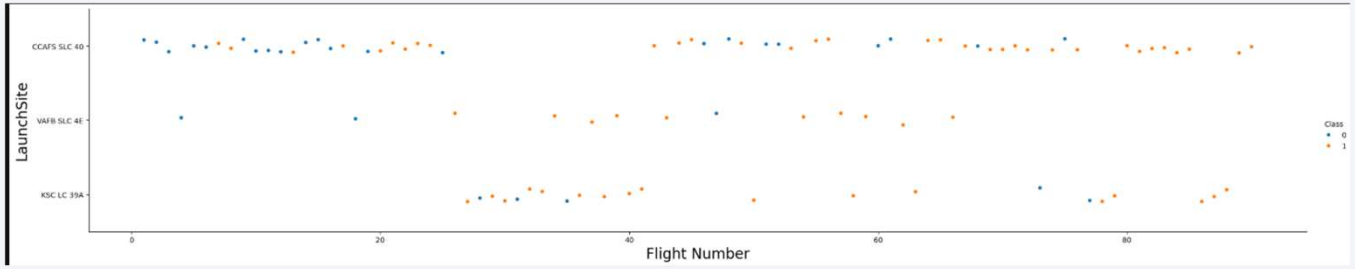
- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results

Section 2

Insights drawn from EDA

Flight Number vs. Launch Site

- Show a scatter plot of Flight Number vs. Launch Site
- Show the screenshot of the scatter plot with explanations

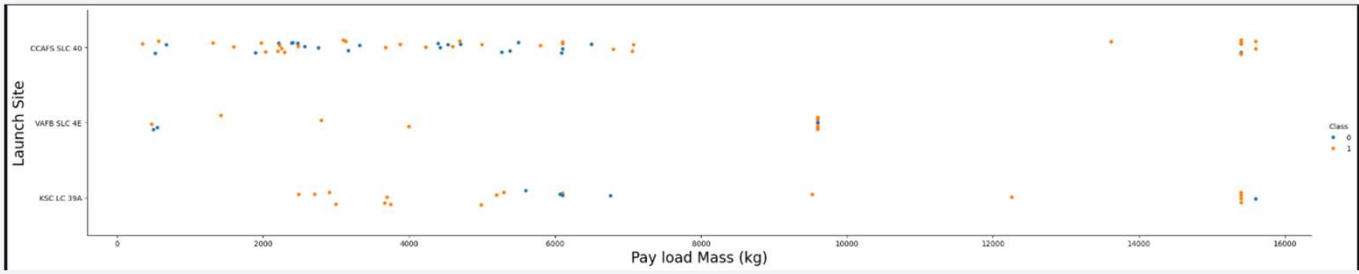


Explanation:
- There are different clustering for each as the Flight Number increases
- Failure decreases as the Flight Number increases for different launch sites

Payload vs. Launch Site

- Show a scatter plot of Payload vs. Launch Site

- Show the screenshot of the scatter plot with explanations

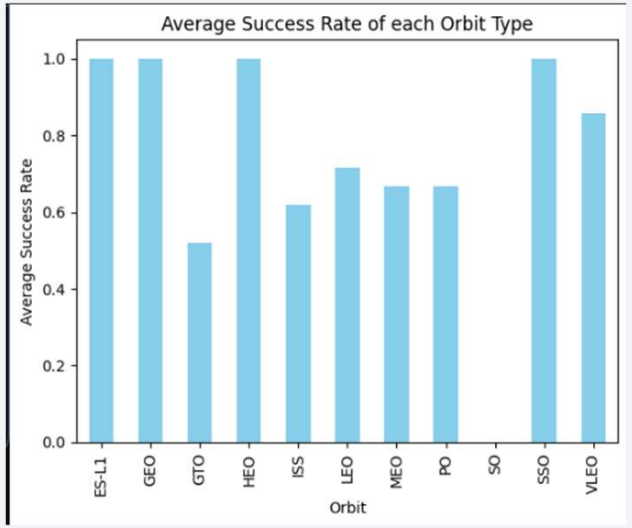


Explanation:
- There is limitation of the payload mass for the launch site, site like VAFB-SLC does not have rocket for certain heavy payload

Success Rate vs. Orbit Type

- Show a bar chart for the success rate of each orbit type

- Show the screenshot of the scatter plot with explanations



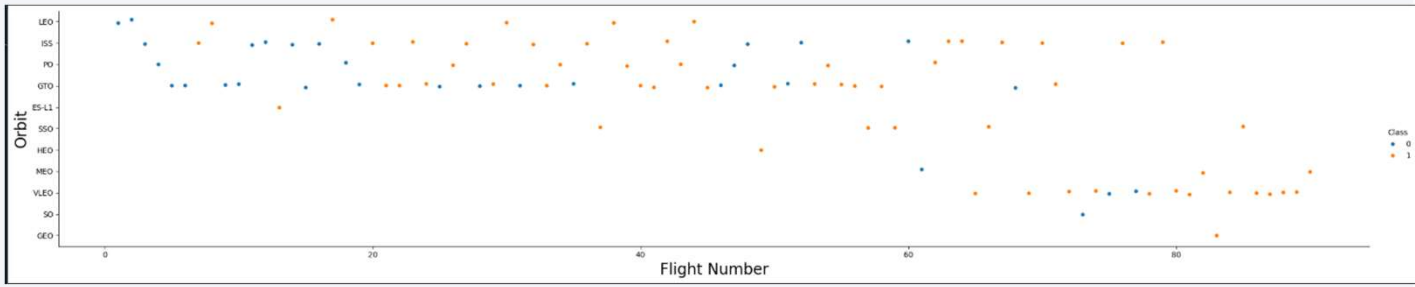
Explanation:

- Orbit of ES-L1, GEO, HEO and SSO have the highest success rate
- Orbit of SO has zero success rate

Flight Number vs. Orbit Type

- Show a scatter point of Flight number vs. Orbit type

- Show the screenshot of the scatter plot with explanations



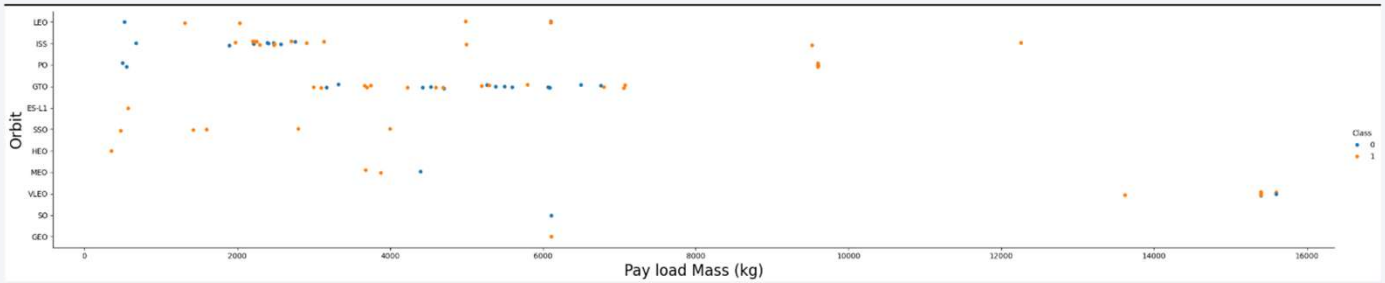
Explanation:

- There is positive relationship for flight number and success rate for some orbit type, e.g. LEO
- There is no relationship for flight number and success rate for some orbit type, e.g. GTO

Payload vs. Orbit Type

- Show a scatter point of payload vs. orbit type

- Show the screenshot of the scatter plot with explanations



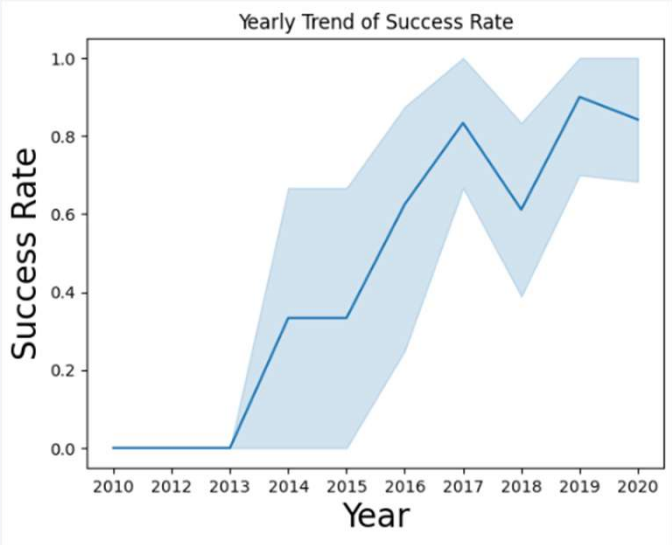
Explanation:

- There is positive relationship for payload mass and success rate for some orbit type, e.g. LEO, PO and ISS
- There is no indicative relationship for payload mass and success rate for some orbit type, e.g. GTO

Launch Success Yearly Trend

- Show a line chart of yearly average success rate

- Show the screenshot of the scatter plot with explanations



Explanation:

- The success rate keeps increasing generally from 2013 to 2020

All Launch Site Names

- Find the names of the unique launch sites
- Present your query result with a short explanation here

```
%sql select distinct Launch_Site from SPACEXTABLE
```

```
 * sqlite:///my_data1.db
Done.
```

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

Explanation:

- The query shows the unique name of launch sites

Launch Site Names Begin with 'CCA'

- Find 5 records where launch sites begin with `CCA`
- Present your query result with a short explanation here

```
%sql select * from SPACEXTABLE where Launch_Site like 'CCA%' limit 5
```

```
 * sqlite:///my_data1.db
Done.
```

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Explanation:

- The query shows only 5 records with the launch sites with the prefix of CCA

Total Payload Mass

- Calculate the total payload carried by boosters from NASA
- Present your query result with a short explanation here

```
%sql select sum("PAYLOAD_MASS__KG_") from SPACEXTABLE where Customer = "NASA (CRS)"
```

```
 * sqlite:///my_data1.db
Done.
```

sum("PAYLOAD_MASS__KG_")
45596

Explanation:

- The query shows the total payload carried by boosters from NASA (CRS) is 45596kg

Average Payload Mass by F9 v1.1

- Calculate the average payload mass carried by booster version F9 v1.1
- Present your query result with a short explanation here

```
%sql select avg(PAYLOAD_MASS__KG_) from SPACEXTABLE where Booster_Version = "F9 v1.1"
 * sqlite:///my_data1.db
Done.
avg(PAYLOAD_MASS__KG_)
                  2928.4
```

Explanation:

- The query shows average payload mass carried by booster version of F9 v1.1 is 2928.4kg

First Successful Ground Landing Date

- Find the dates of the first successful landing outcome on ground pad
- Present your query result with a short explanation here

```
%sql select min(Date) from SPACEXTABLE where Landing_Outcome = "Success (ground pad)"
 * sqlite:///my_data1.db
Done.
```

min(Date)
2015-12-22

Explanation:

- The query shows the first successful landing outcome of ground pad is 2015-12-22

Successful Drone Ship Landing with Payload between 4000 and 6000

- List the names of boosters which have successfully landed on drone ship and had payload mass greater than 4000 but less than 6000

- Present your query result with a short explanation here

```
%sql select distinct Customer from SPACEXTABLE where Landing_Outcome = "Success (drone ship)" and PAYLOAD_MASS__KG_ > 4000 and PAYLOAD_MASS__KG_ < 6000
```

```
 * sqlite:///my_data1.db
Done.
```

Customer
SKY Perfect JSAT Group
SES
SES EchoStar

Explanation:

- The query shows the names of successful drone ship landing with payload mass between 4000 and 6000 are SES and SEC EchoStar

Total Number of Successful and Failure Mission Outcomes

- Calculate the total number of successful and failure mission outcomes
- Present your query result with a short explanation here

```
%sql select Mission_Outcome, count(1) from SPACEXTABLE group by Mission_Outcome
```

```
 * sqlite:///my_data1.db
Done.
```

Mission_Outcome	count(1)
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

Explanation:

- The query shows the total number of successful and failure mission outcomes

Boosters Carried Maximum Payload

- List the names of the booster which have carried the maximum payload mass
- Present your query result with a short explanation here

```
%sql select distinct Booster_Version from SPACEXTABLE where PAYLOAD_MASS__KG_ = (select max(PAYLOAD_MASS__KG_) from SPACEXTABLE)
 * sqlite:///my_data1.db
Done.
```

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

Explanation:

- The query shows the list of boosters that carried maximum payload mass

2015 Launch Records

- List the failed landing_outcomes in drone ship, their booster versions, and launch site names for in year 2015

- Present your query result with a short explanation here

```
%sql select substr(Date, 6, 2) as Month, Landing_Outcome, Booster_Version, Launch_Site from SPACEXTABLE where Landing_Outcome = "Failure (drone ship)" and substr(Date, 0, 5) = '2015'
```

```
 * sqlite:///my_data1.db
Done.
```

Month	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

Explanation:

- The query shows the list of month, failed landing outcome in drone ship, booster versions and launch site in year 2015

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order
- Present your query result with a short explanation here

```
%sql select Landing_Outcome, count(1) as Landing_Count from SPACEXTABLE where Date >=  "2010-06-04 " and Date <= "2017-03-20" group by Landing_Outcome order by Landing_Count desc
 * sqlite:///my_data1.db
Done.
```

Landing_Outcome	Landing_Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Precluded (drone ship)	1
Failure (parachute)	1

Explanation:

- The query shows the list of rank landing outcome between 2010-06-04 and 2017-03-20

Section 3

Launch Sites Proximities Analysis

<Folium Map Screenshot 1>

- Replace <Folium map screenshot 1> title with an appropriate title
- Explore the generated folium map and make a proper screenshot to include all launch sites' location markers on a global map
- Explain the important elements and findings on the screenshot

<Folium Map Screenshot 2>

- Replace <Folium map screenshot 2> title with an appropriate title
- Explore the folium map and make a proper screenshot to show the color-labeled launch outcomes on the map
- Explain the important elements and findings on the screenshot

<Folium Map Screenshot 3>

- Replace <Folium map screenshot 3> title with an appropriate title

- Explore the generated folium map and show the screenshot of a selected launch site to its proximities such as railway, highway, coastline, with distance calculated and displayed

- Explain the important elements and findings on the screenshot

Section 4

Build a Dashboard with Plotly Dash

<Dashboard Screenshot 1>

- Replace <Dashboard screenshot 1> title with an appropriate title
- Show the screenshot of launch success count for all sites, in a piechart
- Explain the important elements and findings on the screenshot

<Dashboard Screenshot 2>

- Replace <Dashboard screenshot 2> title with an appropriate title
- Show the screenshot of the piechart for the launch site with highest launch success ratio
- Explain the important elements and findings on the screenshot

<Dashboard Screenshot 3>

- Replace <Dashboard screenshot 3> title with an appropriate title
- Show screenshots of Payload vs. Launch Outcome scatter plot for all sites, with different payload selected in the range slider
- Explain the important elements and findings on the screenshot, such as which payload range or booster version have the largest success rate, etc.

Section 5

Predictive Analysis (Classification)

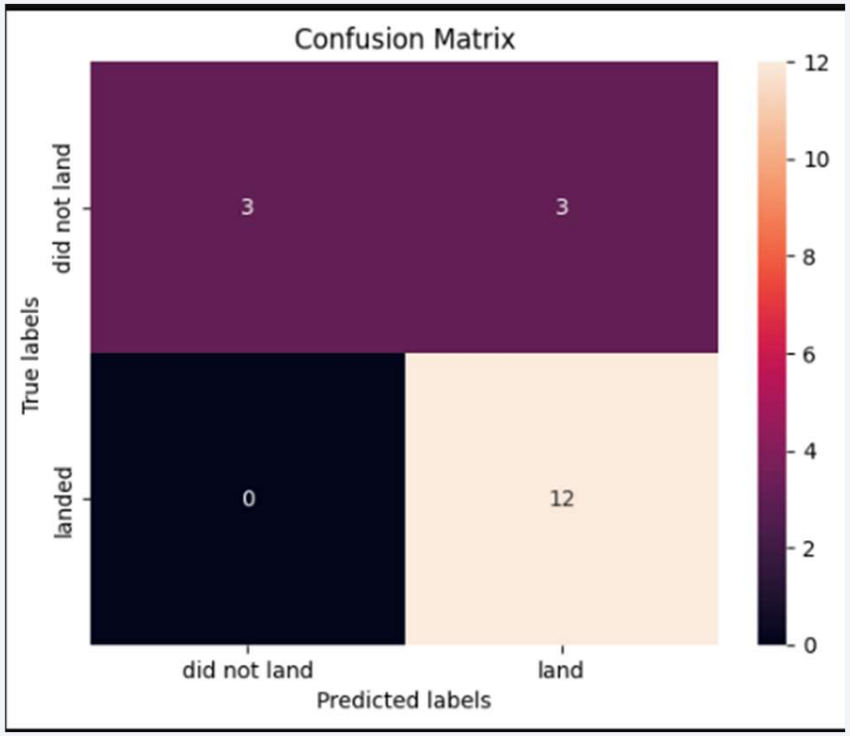
Classification Accuracy

- Visualize the built model accuracy for all built classification models, in a bar chart

- Find which model has the highest classification accuracy

Confusion Matrix

- Show the confusion matrix of the best performing model with an explanation



Conclusions

- The public data can be wrangled for analysis
- The correlation between different factor and success or failure rate is identified
- The impact of different proximity on the success or failure rate is identified
- The highest performing classification model is identified
- The objective to determine the impact of first launch to the cost is achieved

Appendix

- Include any relevant assets like Python code snippets, SQL queries, charts, Notebook outputs, or data sets that you may have created during this project

Thank you!